

Foundation of Blockchain Technology (IS501) Pre-Reading Material – Lecture 12

Introduction to Ethereum

Estimated Reading Time: 15–20 minutes

Why This Topic Matters

Bitcoin demonstrated how blockchain can be used for decentralized digital currency. However, its programmability is limited.

Ethereum extended the idea of blockchain by introducing a programmable platform for **smart contracts and decentralized applications (DApps)**. This development is commonly associated with **Blockchain 2.0**.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Explain why Ethereum is called **Blockchain 2.0**.
- Understand the concept of **smart contracts**.
- Explain the role of **DApps**.
- Identify major applications of Ethereum.
- Understand the basic idea of **DeFi**.
- Explain the use of Ethereum in **digital identity**.

Activate Your Prior Knowledge

1. Bitcoin

Bitcoin uses blockchain for decentralized digital transactions.

Think:

Can blockchain be used for something more than transferring cryptocurrency?

2. Blockchain Transactions

Blockchain records transactions in a verifiable and tamper-resistant manner.



Think:

What if a blockchain could automatically execute an action when a condition is satisfied?

3. Decentralization

Blockchain reduces dependence on central authorities.

Think:

Can software running on a blockchain replace some functions normally performed by intermediaries?

Prerequisite Concepts

You should be familiar with:

- Blockchain
- Bitcoin
- Cryptocurrency
- Decentralization
- Blockchain transactions
- Smart-contract basics

Core Concepts

1. What is Ethereum?

Ethereum is a programmable blockchain platform introduced by **Vitalik Buterin** in **2013**.

Unlike Bitcoin, which was primarily designed for digital currency, Ethereum was designed to support **smart contracts and decentralized applications**.

Bitcoin vs Ethereum

Bitcoin

Ethereum

Primarily digital currency Programmable blockchain

Limited programmability Smart contracts

Focus on value transfer Supports DApps and multiple applications



2. Why is Ethereum Called Blockchain 2.0?

The evolution can be viewed as:

Blockchain 1.0 → Bitcoin → Digital Currency

Blockchain 2.0 → Ethereum → Programmable Blockchain

The key idea is that blockchain can now be used not only to record transactions but also to **execute programmed rules and applications**.

3. Smart Contracts

A **smart contract** is code stored on a blockchain that automatically executes predefined actions when specified conditions are satisfied.

A simple example:

IF loan is fully repaid

↓

Release collateral

Another example:

IF goods are delivered and verified

↓

Release payment

These examples show how smart contracts can automate agreements.

4. Decentralized Applications (DApps)

Smart contracts allow developers to build **Decentralized Applications (DApps)**.

A simple relationship is:

Ethereum → Smart Contracts → DApps



DApps can be used in areas such as:

- Finance
- Supply chain
- Real estate
- Digital identity

5. Ethereum in Financial Services

Ethereum can provide programmable infrastructure for financial applications.

Important advantages include:

Transparency

Transactions can be publicly verified and ownership history can be tracked.

Disintermediation

Blockchain can reduce dependence on intermediaries, potentially reducing costs and settlement time.

Automation

Smart contracts can automate processes such as loan approvals and escrow releases.

6. Decentralized Finance (DeFi)

DeFi refers to financial applications built using decentralized blockchain infrastructure.

Examples include:

- **Lending and Borrowing** – peer-to-peer financial services.
- **Stablecoins** – crypto assets designed to maintain relatively stable value.
- **Decentralized Exchanges (DEXs)** – platforms for decentralized asset swaps.

7. Ethereum and Digital Identity

Ethereum can also be used for **digital identity management**.

The idea is to associate blockchain accounts with verifiable real-world claims.



The source material mentions **ERC-725** as a standard for associating Ethereum accounts with real-world claims.

Potential advantages include:

- **User control** over identity information
- **Security**
- **Interoperability**

Possible applications include:

- KYC and compliance
- Cross-border identity
- IoT device authentication

Guided Reading Questions

1. Why is Ethereum considered **Blockchain 2.0**?
2. What is a **smart contract**?
3. Give one real-world example of a smart contract.
4. What are **DApps**?
5. What is meant by **disintermediation**?
6. What is **DeFi**?
7. Name three DeFi applications.
8. How can Ethereum be used for digital identity?
9. What is the role of **ERC-725**?
10. How is Ethereum different from Bitcoin?

Reflection Activity

Write 75–100 words explaining how Ethereum extends blockchain technology beyond cryptocurrency. Mention **smart contracts** and at least **two applications**.

Self-Assessment

1. Ethereum is commonly associated with Blockchain _____.
2. Ethereum was introduced by _____.
3. A blockchain-based program that automatically executes predefined rules is called a _____.
4. Decentralized applications are called _____.
5. DeFi stands for Decentralized _____.
6. A decentralized exchange is commonly abbreviated as _____.
7. Ethereum can be used for digital _____ management.

8. ERC-725 is related to blockchain-based _____.

Key Takeaway

Ethereum transformed blockchain from a system primarily used for digital currency into a programmable platform.

The key concepts to remember are:

Ethereum → Smart Contracts → DApps → Real-World Applications

Major applications include **financial services, DeFi, real estate, supply chains, and digital identity.**

